

Exercise 1E Solutions

Word Problems Involving Integers

Comprehensive Fundamentals of Integer Mathematics

1. Theoretical Foundations

Solving word problems with integers requires translating real-world scenarios into structured mathematical equations using signed values. Below are key arithmetic and conceptual principles:

Mathematical Translation Rules

- **Difference/Exceedance:** To determine how much a value a exceeds b , compute the difference:

$$\text{Exceedance} = a - b$$

- **Deficit/Comparison:** To find how much a value a is less than b , compute:

$$\text{Deficit} = b - a$$

- **Subtractive Unknowns:** If a value x is subtracted from a given number p to yield q , the algebraic setup is:

$$p - x = q \implies x = p - q$$

- **Signed Vector Quantities:**

- *Direction:* East is represented as positive (+), and West as negative (–).
- *Temperature:* Temperatures above 0°C are positive, while temperatures below 0°C are negative.

- **Financial Transactions:** Revenue or Selling Price (SP) is positive, whereas Cost Price (CP) or Expenditure is negative. A net positive balance indicates a **Gain**, whereas a net negative balance indicates a **Loss**.

2. Step-by-Step Solutions

Question 1

Problem: By how much does 5 exceed -5 ?

Step-by-Step Solution:

1. To find by how much an integer a exceeds another integer b , we subtract b from a :

$$\text{Exceedance} = a - b$$

2. Let $a = 5$ and $b = -5$. Substitute these values into the formula:

$$\text{Exceedance} = 5 - (-5)$$

3. By the rule of subtraction (subtracting a negative integer is equivalent to adding its additive inverse):

$$5 - (-5) = 5 + 5 = 10$$

Final Answer: 5 exceeds -5 by **10**.

Question 2

Problem: How much is -7 less than -1 ?

Step-by-Step Solution:

1. To find how much an integer a is less than another integer b , we subtract a from b :

$$\text{Deficit} = b - a$$

2. Let $a = -7$ and $b = -1$. Substitute these values:

$$\text{Deficit} = -1 - (-7)$$

3. Since subtracting -7 is equivalent to adding its additive inverse 7 :

$$-1 - (-7) = -1 + 7$$

4. Since the integers have unlike signs, we find the difference of their absolute values ($7 - 1 = 6$) and give it the sign of the larger absolute value (which is positive):

$$-1 + 7 = 6$$

Final Answer: -7 is less than -1 by **6**.

Question 3

Problem: What must be subtracted from 7 to get -6 ?

Step-by-Step Solution:

1. Let the unknown integer to be subtracted be x .
2. Formulate the algebraic equation from the problem description:

$$7 - x = -6$$

3. Transpose x to the right-hand side and -6 to the left-hand side to solve for x :

$$x = 7 - (-6)$$

4. Convert the subtraction of a negative number into addition:

$$x = 7 + 6 = 13$$

Final Answer: The integer that must be subtracted is **13**.

Question 4

Problem: What must be subtracted from -1 to get -19 ?

Step-by-Step Solution:

1. Let the unknown integer to be subtracted be x .

2. Write the algebraic equation:

$$-1 - x = -19$$

3. Solve for x by rearranging the terms:

$$x = -1 - (-19)$$

4. Apply the sign rule for subtracting a negative number:

$$x = -1 + 19$$

5. Find the difference of the absolute values ($19 - 1 = 18$). Since 19 is positive and has the larger absolute value, the result is positive:

$$x = 18$$

Final Answer: The integer that must be subtracted is **18**.

Question 5

Problem: The sum of two integers is -11 . If one of them is 5, find the other.

Step-by-Step Solution:

1. Let the unknown integer be y .

2. The sum of the two integers is given as -11 , and one of them is 5:

$$y + 5 = -11$$

3. Isolate y by subtracting 5 from both sides of the equation:

$$y = -11 - 5$$

4. Since both integers being added have like negative signs (adding -11 and -5):

$$y = -(11 + 5) = -16$$

Final Answer: The other integer is -16 .

Question 6

Problem: Add the product of (-13) and (-17) to the quotient of (-187) and 11 .

Step-by-Step Solution:

1. **Calculate the product:** Multiply (-13) and (-17) . Since both integers have the same negative sign, the product is positive:

$$\text{Product} = (-13) \times (-17) = +(13 \times 17)$$

$$13 \times 17 = 13 \times (10 + 7) = 130 + 91 = 221$$

2. **Calculate the quotient:** Divide (-187) by 11 . Since the dividend and the divisor have unlike signs, the quotient is negative:

$$\text{Quotient} = (-187) \div 11 = -17$$

3. **Perform the final addition:** Add the product to the quotient:

$$\text{Result} = 221 + (-17) = 221 - 17 = 204$$

Final Answer: The final evaluated result is **204**.

Question 7

Problem: A shopkeeper bought a pen for Rs. 75, a book for Rs. 240 and a pencil box for Rs. 46. He sold the pen for Rs. 81, book for Rs. 255 and the pencil box for Rs. 40. What was his gain or loss?

Step-by-Step Solution:

1. **Calculate the total Cost Price (CP):** Add the cost price of all items bought:

$$CP = 75 + 240 + 46$$

$$CP = 315 + 46 = 361$$

Total Cost Price = Rs. 361.

2. **Calculate the total Selling Price (SP):** Add the selling price of all items sold:

$$SP = 81 + 255 + 40$$

$$SP = 336 + 40 = 376$$

Total Selling Price = Rs. 376.

3. **Compare SP and CP:** Since the Selling Price is greater than the Cost Price ($SP > CP$), the shopkeeper made a **Gain**:

$$\text{Gain} = SP - CP$$

$$\text{Gain} = 376 - 361 = 15$$

Final Answer: The shopkeeper made a net **Gain of Rs. 15**.

Question 8

Problem: A man starts from his home and drives 169 km to the East and then 192 km to the West. How far is he from his home finally and in which direction?

Step-by-Step Solution:

1. Let the man's home be the reference point (0 on a horizontal number line).
2. Let driving to the **East** represent the positive direction (+) and driving to the **West** represent the negative direction (-).
3. Express the movements as integers:
 - Driving 169 km to the East = +169 km
 - Driving 192 km to the West = -192 km
4. Find his final position relative to home by adding the movements:

$$\text{Final Position} = +169 + (-192) = 169 - 192 = -23 \text{ km}$$

5. The negative sign represents the direction **West**. The absolute value representing distance is $|-23| = 23$.

Final Answer: He is finally **23 km to the West** from his home.

Question 9

Problem: During a month, the average day temperature of a desert region was 43°C and the average night temperature was -6°C . Find the difference between the average day temperature and the average night temperature.

Step-by-Step Solution:

1. Let $T_{\text{day}} = 43^{\circ}\text{C}$ and $T_{\text{night}} = -6^{\circ}\text{C}$.
2. To find the difference between the two temperatures, subtract the lower temperature (night) from the higher temperature (day):

$$\text{Difference} = T_{\text{day}} - T_{\text{night}}$$

$$\text{Difference} = 43^{\circ}\text{C} - (-6^{\circ}\text{C})$$

3. Apply subtraction sign rules:

$$\text{Difference} = 43 + 6 = 49^{\circ}\text{C}$$

Final Answer: The difference between the day and night temperatures is 49°C .

Question 10

Problem: Radhey bought 4 pairs of jeans at Rs. 1256 each. How much did he pay for the jeans in all?

Step-by-Step Solution:

1. Cost of a single pair of jeans = Rs. 1256.
2. Number of pairs bought = 4.
3. To find the total cost, multiply the unit cost by the quantity:

$$\text{Total Paid} = 4 \times 1256$$

4. Break down the multiplication:

$$4 \times 1256 = 4 \times (1000 + 200 + 50 + 6)$$

$$= 4000 + 800 + 200 + 24 = 5024$$

Final Answer: Radhey paid a total of **Rs. 5024** for the jeans.

Question 11

Problem: Mathew had no money to pay for his house rent of Rs.14240. His 5 friends decided to contribute equally to pay for the rent. How much will each friend pay?

Step-by-Step Solution:

1. Total rent amount to be paid = Rs.14240.
2. Number of contributing friends = 5.
3. To find the individual contribution of each friend, divide the total rent by the number of friends:

$$\text{Contribution} = 14240 \div 5$$

4. Perform the division:

$$14240 \div 5 = 2848$$

Final Answer: Each friend will pay **Rs. 2848**.

Question 12

Problem: A class of 36 students contributed Rs. 540 each for the picnic. Out of these, 7 students decided not to go. They were returned Rs. 490 each. What amount was spent on the picnic?

Step-by-Step Solution:

1. **Calculate total amount collected initially:** Multiplying the number of students (36) by the contribution of each student (Rs. 540):

$$\text{Total Collected} = 36 \times 540$$

$$36 \times 540 = 19440$$

Initial collection = Rs. 19440.

2. **Calculate total amount returned to the 7 students:** Multiplying the number of non-attending students (7) by the returned amount (Rs. 490):

$$\text{Total Returned} = 7 \times 490$$

$$7 \times 490 = 3430$$

Total refund = Rs. 3430.

3. **Calculate the net amount spent on the picnic:** Subtract the returned amount from the initially collected amount:

$$\text{Amount Spent} = \text{Total Collected} - \text{Total Returned}$$

$$\text{Amount Spent} = 19440 - 3430 = 16010$$

Final Answer: The total amount spent on the picnic was **Rs. 16010**.